

IN-SITU RELATIVE HUMIDITY TEST REPORT

Report No.: CMT-11-0412

Date of report: February 8, 2011

Project: 632588 — Ardsley Financial Group Headquarters, Corbin Tower, Levels 12–19

Client: VAA interiors + architecture

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Test method: ASTM F2170 — in-situ probe, 40% slab depth, sealed, read at 72 hours

Specification threshold: 85% RH (Section 09 05 61, Article 1.2.C)

1. Substrate

Lightweight structural concrete fill, nominal 3-1/4 inches, over 3 inch composite metal deck. Placed between June and September 2010. Building enclosed and conditioned since December 2010. Slab dries from the upper face only.

2. Results

This is the **preliminary screening series** requested by the Architect to establish the general moisture condition of the slab ahead of the full compliance series required by Section 09 05 61 Article 1.2.B. It is not, and does not purport to be, the full-density sampling that Article requires prior to installation.

Probes installed February 1, 2011. Read February 4, 2011. Ambient 71°F / 44% RH at time of reading.

Probe	Level	Location	RH (%)	Slab temp (°F)
P-01	12	Northeast core	92.1	68.9
P-02	12	West perimeter, grid C-4	94.6	68.4
P-03	13	Center bay, grid E-6	95.2	68.1
P-04	13	South perimeter, grid B-9	93.8	68.6
P-05	14	Center bay, grid E-6	95.9	68.0
P-06	14	Pantry, grid H-2	96.4	67.8
P-07	15	Northeast core	93.0	68.7
P-08	15	Center bay, grid E-6	95.5	68.2
P-09	16	Center bay, grid E-6	96.1	68.0
P-10	16	West perimeter, grid C-4	94.2	68.5
P-11	17	Center bay, grid E-6	95.7	68.1
P-12	17	South perimeter, grid B-9	92.4	68.8
P-13	18	Center bay, grid E-6	95.0	68.3

Probe	Level	Location	RH (%)	Slab temp (°F)
P-14	19	Pantry and wellness, grid H-2	96.0	67.9

Range: 92.1% to 96.4% RH. Mean: 94.7%. Twelve of fourteen probes exceed 93% RH. No probe met the 85% specification threshold.

3. Observations

The lowest reading recorded on the eight floors was 92.1% at the Level 12 northeast core, the earliest slab placed and the area longest under conditioning. The distribution is consistent with a lightweight fill over metal deck of this age and thickness.

4. Conclusion

No screened area on Levels 12 through 19 approaches the acceptance threshold stated in Section 09 05 61, Article 1.2.C, and no result is close enough to it that full-density sampling could plausibly reverse the conclusion. In this laboratory's opinion the slab will not reach 85% RH on any schedule compatible with the construction program. Moisture vapor mitigation is required prior to the installation of adhered floor coverings.

Retest following mitigation is recommended and is required by the specification.

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